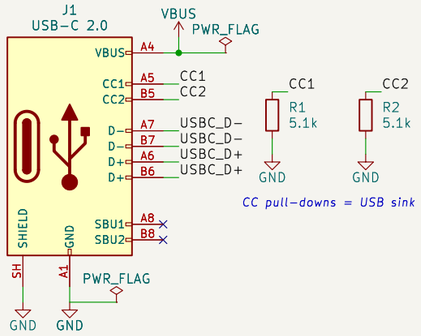
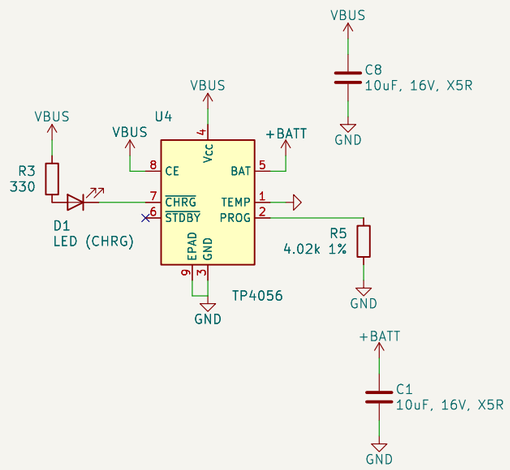
USB-C INPUT

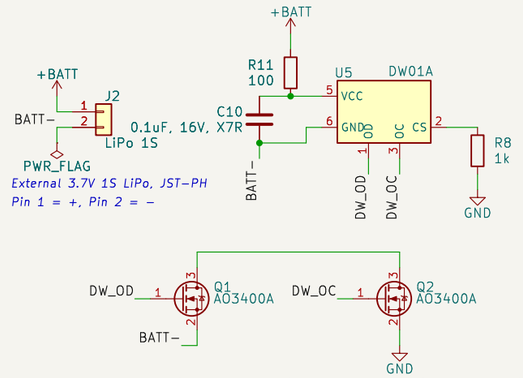


CC pull-downs = USB sink

LI-ION CHARGER (TP4056)

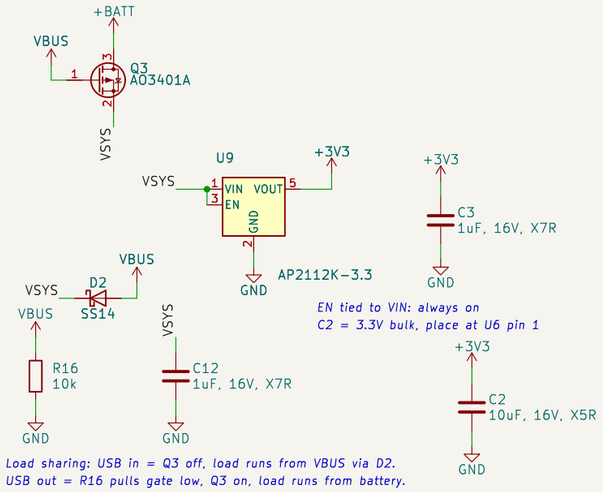


BATTERY + PROTECTION (DW01A)



Q1 = discharge FET (source on BATT-), Q2 = charge FET (source on GND).
AO3400A, SOT-23: pin 1 G, 2 S, 3 D. R11/C10 = DW01A VCC filter.

LOAD SHARING + 3.3V REGULATOR



EN tied to VIN: always on
C2 = 3.3V bulk, place at U6 pin 1

Load sharing: USB in = Q3 off, load runs from VBUS via D2.
USB out = R16 pulls gate low, Q3 on, load runs from battery.

MCU (ESP32-C3-WROOM-02)

Hold BOOT while resetting = download mode

R14/R15: strapping pull-ups

R12, R13 near U6. C2 + C5 at U6 pin 1. IO5 = accel wake, IO3 = RTC alarm wake

I2C BUS + OLED

I2C: RTC 0x68, EEPROM 0x50,
ADXL343 0x53, OLED 0x3C

RTC (DS3231)

Fit BT2: VBAT must not float
R17: INT/SQW is open drain (alarm wakes ESP32 on IO3)

EEPROM (24LC256)

A0-A2 = GND: address 0x50

ACCELEROMETER (ADXL343)

CS high = I2C mode, SDO low = 0x53. INT1 -> IO5 (wake on motion). C13 at pin 6, C6 at pin 1

HALL SENSOR

U7 = DRV5032FADBZR, SOT-23: 1 VCC, 2 OUT, 3 GND. Push-pull, so R4 not fitted.
Senses field through the chip face. Magnet must face U7, within 12 mm.
OUT low = magnet near. C11 = VCC decoupling.

Mounting holes: M2, 2.2 mm

H1 M2 hole
H2 M2 hole
H3 M2 hole

Rev 2: fixes CE, charge LED, Hall pinout, protection FETs; adds load sharing, wake interrupts

Veer Sharma

Sheet: /
File: MedicineStorage.kicad_sch

Title: Medication Adherence Monitor

Size: A3	Date: 2026-09-28	**Rev: 2.0 (fixes)**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1